What is Computer network?

"1. When a computer is connected to another computer, we call it a network.

2. In simple words, "connecting two or more computers to a computer is called a network".

3. Through the network, computers share data and information with each other.

4. A network can be created both via wire and wireless.

5. The network has such a high capacity to connect computers that devices can be connected from one corner to every corner of the world."

Data communication

Data communications are the exchange of data between two devices via some transmission medium.

The effectiveness of a data communications system depends on four fundamental characteristics: delivery, accuracy, timeliness, and jitter. I. Delivery. The system must deliver data to the correct destination. Data must be received by the intended device or user and only by that device or user. 2 Accuracy. The system must deliver the data accurately. Data that have been altered in transmission and left uncorrected are unusable. 3. Timeliness. The system must deliver data in a timely manner. Data delivered late are useless. In the case of video and audio, timely delivery means delivering data as they are produced, in the same order that they are produced, and without significant delay. This kind of delivery is called real-time transmission. 4. Jitter. Jitter refers to the variation in the packet arrival time. It is the uneven delay in the delivery of audio or video packets. For example, let us assume that video packets are sent every 30 ms. If some of the packets arrive with 30-ms delay and others with 40-ms delay, an uneven quality in the video is the result.

A data communications system has five components

1.Message. The message is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

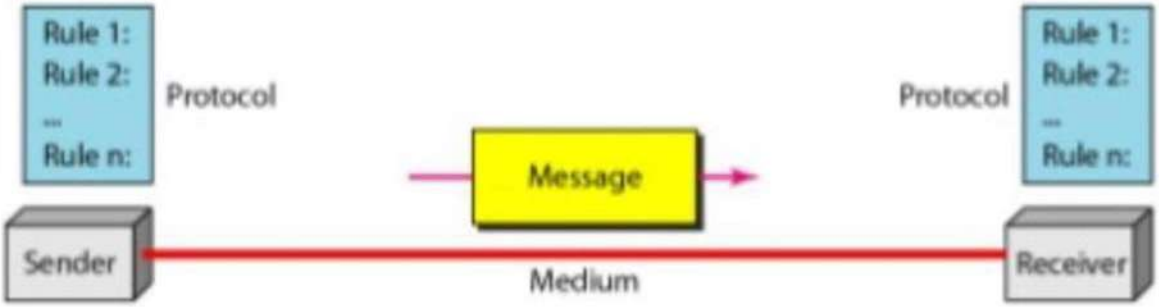
2 Sender. The sender is the device that sends the data message. It can be a computer, workstation, telephone handset, video camera, and so on.

3. Receiver. The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on

. **4. Transmission medium.** The transmission medium is the physical path by which a message travels from sender to receiver. Some examples of transmission media include twisted-pair wire, coaxial cable,

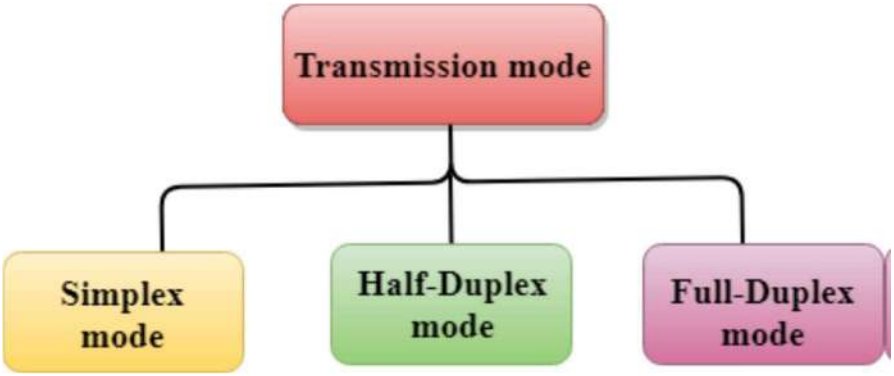
fiber-optic cable, and radio waves.

5. Protocol. A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.



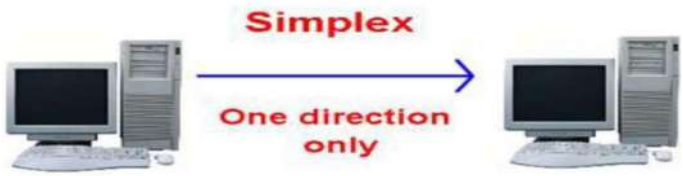
Transmission Mode

Data flow between two systems can be categorised into three types –



Simplex

The communication is unidirectional as a one-way street, one device always, other can always receive. E.g. radio, mouse. The simplex mode can use the entire capacity of the channel to send data in one direction.



Half duplex

Each station can both transmit and receive, but not at the same time. E.g. like a one lane road, walkie-talkie etc.When one device is sending, the other can only receive, and vice versa.

In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.

Walkie-talkies are both half-duplex systems.



Walkie-talkies

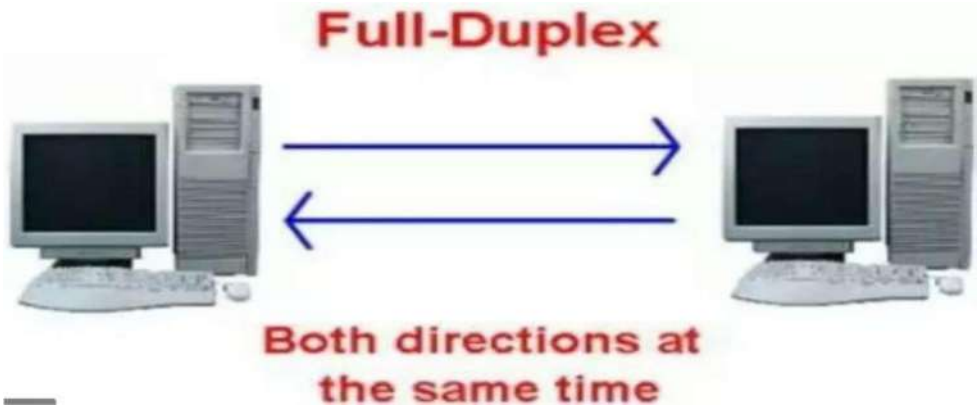


Full duplex

Both stations can transmit and receive at the same time. Actually, it is two half duplex connections.

Telephone network is an example of full-duplex mode, when two people are communicating by a telephone line, both can talk and listen at the same time.

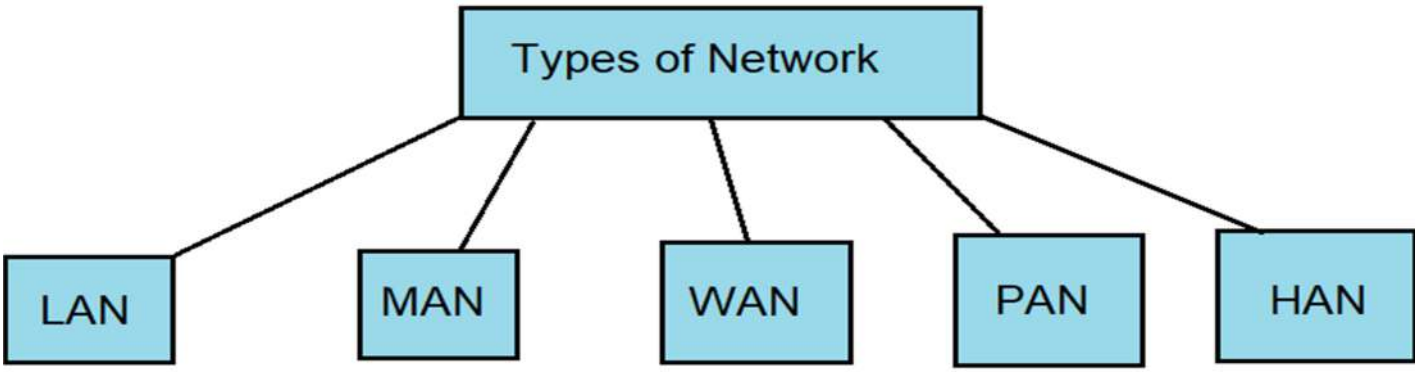
The capacity of the channel, must be divided between the two directions.



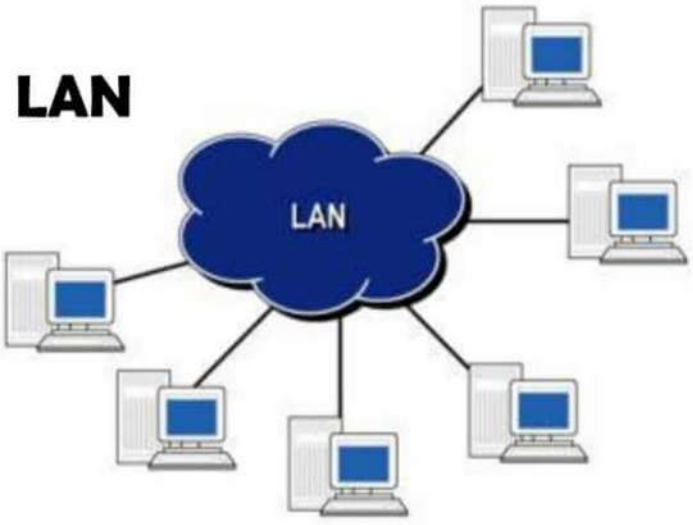
Types Of Network

"There are following 5 types of networks:- LAN

1. MAN
2. WAN
3. PAN
4. HAN



LAN (Local area network)



"• The full form of LAN is Local Area Network.

• LAN is a network that is used to connect computers or devices in a small area. Such as – in office, school, college etc.

• We can see such networks in schools, colleges, offices etc.

• Creating a LAN is very cheap compared to other networks.

• LAN is used to share data, store data and print documents.

• Very big software is not used to create a LAN. Hub, switch, network adapter, router and Ethernet cable are used to connect computers in LAN.

• The speed of data transfer in it is very high."

Advantages of LAN

"Its advantages are given below.

1. In this, data can be shared from one computer to another very easily.

2. Through this, we can also share the Internet.

3. Through this, software programs can be shared.

4. In this, communication is very easy and fast.

5. LAN saves time.

6. It is very easy to create a LAN.

7. In Local Area Network, We can connect more than 1000 computers to each other at a time."

Disadvantages of LAN

Its disadvantages are as follows-

1. There is a risk of virus or malware spreading in this network.

2. The area of LAN is very small, so we can share data only in a small area.

3. There is a possibility of server area crashing.

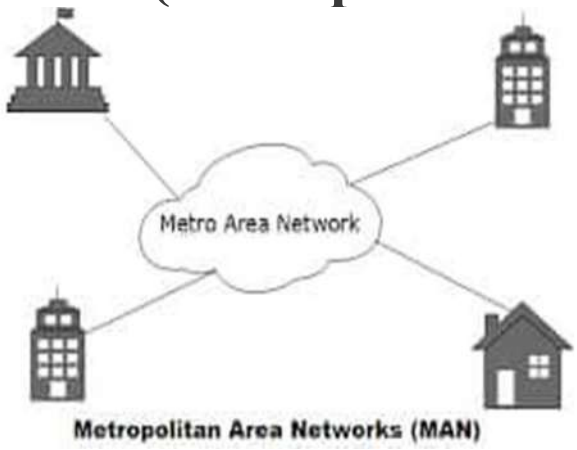
4. It is very difficult to maintain LAN.

5. The security in it is not good. That means it is easy to hack it.

6. It is ve ry difficult to maintain such a network.

7. It is difficult to set it up."

MAN (Metropolitan Area Network)



"• The full form of MAN is Metropolitan Area Network.

• The area of MAN is bigger than LAN but smaller than WAN.

• MAN network is like LAN but the area of MAN is bigger.

• MAN is a network that connects many computers in a city.

For example – Suppose We have many computers at different places in Delhi. Now We want all these computers to be connected to one network. In such a situation We will need a MAN network. That is, a network that can connect many computers from one place.

• The range of MAN network is around 10 KM to 100Km. This means that We can cover an area of 100 km with the help of one network. Which is a lot in itself. MAN network is used only to cover a larger area. The best example of MAN network is cable network. You must have often seen cable network inside the TV in your homes.
• A MAN network is created by an organization or firm itself. Because the firm or organization itself creates this network to connect its branches. The owner of this network is not a single person. Because the ownership of this network is both public and private. For example, MAN is used to connect all the branches of a bank.

• The protocols used in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, and ADSL etc."

Advantages of MAN

"Its advantages are as follows-

1. It covers a longer distance.

2. It is less expensive than LAN.

3. With the help of this network, We can share emails and programs.

4. The speed of this network is quite good.

5. With the help of this network, We can share the internet.

6. LAN can be easily converted into MAN network.

7. We get very good security in this."

Disadvantages of MAN

“Its disadvantages are as follows-

1. It is very difficult to manage this network.

2. Its internet speed can vary.

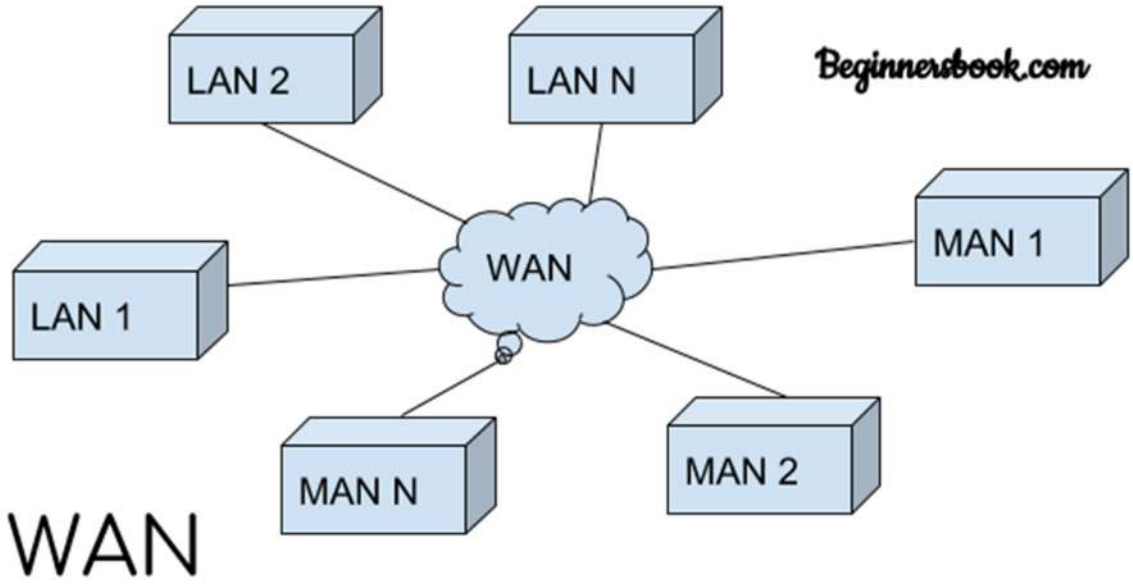
3. Despite having security, it is attacked by hackers.

4. Setting it up is a very difficult task.

5. Technical people are required to setup MAN network.

6. This network requires more wire.”

WAN (Wide Area Network)



“• The full form of WAN is Wide Area Network.

• WAN is the network that connects every computer and device in the world. We also call it international network. Because this network is for state to state and country to country connectivity. Its best example is internet. We can access internet from any corner of the world.

• This network is used to connect one country to the network of another country. It is also considered the largest network in the world. Which connects computers all over the world. This network is also known as LAN of LANs.

• The biggest feature of this network is that it covers a large area. Apart from this, its data rate is also high. WAN networks are also of two types. 1.Enterprise WAN and 2.Global WAN

• Most of the computers connecting to WAN network use public network. • It costs a lot of money to install this network. Apart from this, it is also very difficult to maintain it. The companies which provide this network are called Network Service Providers.

• The area of WAN is much larger than LAN and MAN.
• Wide area network is used more in the field of business, education and government."

Advantages of WAN

"Its advantages are given below.

1. This network covers a very large area.

2. It maintains a centralized data system.

3. Both data and files remain updated in it.

4. You can share many applications and messages in this network.

5. You can easily share software and resources in this network.

6. WAN networks are beneficial for global business.

7. The workload is easily distributed in this network.

8. Its bandwidth is very high."

Disadvantages of WAN

"Its disadvantages are as follows-

1. It is difficult to run this network without antivirus and firewall.

2. It costs a lot to set up this network.

3. There is a problem of troubleshooting in WAN network.

4. There is a problem of server down in this network.

5. It is very difficult to maintain this network."

PAN (Personal Area Network)



"• The full form of PAN is Personal Area Network.

• Personal Area Network is a network which is already connected with the devices.

• The range of this network is very small. This network can be used by only one person. Its range is up to 10 meters only.

• Only personal computers are connected through PAN, hence it is called personal area network.

• This network allows only a few devices. Like computer, smartphone, smartwatch etc.

Advantages of PAN

“Its benefits are as follows.

1. There is no need to connect to any separate network.

2. No one other than you can use this network.

3. This network can transfer any kind of data.

4. It is very easy to use this network.

5. It does not cost much to set up this network.”

Disadvantages of PAN

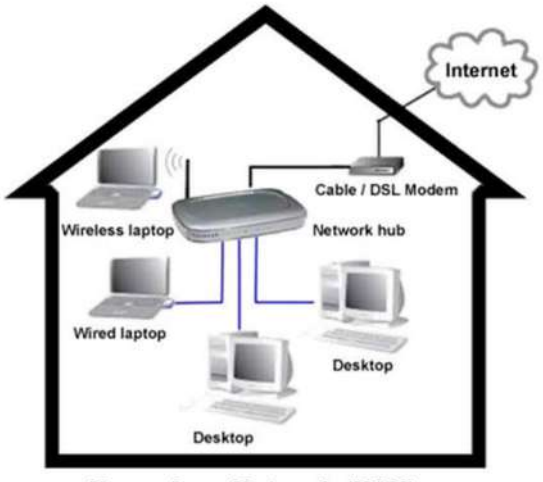
“Its disadvantages are given below.

1. The range of PAN network is very less.

2. The speed of data transfer in it is very slow.

3. These networks crash when a weak signal comes.”

HAN (Home Area Network)



Home Area Network (HAN)

“• The full form of HAN is “Home Area Network”.

• If all the people in the same house use the same network, then we call it HAN network.

• These networks are not personal at all because they are not used by a single person.

• Through this network, we can connect all the devices of the house such as computer, laptop, smart phone etc.

• The range of this network is less than 100 meters.

• Modem and router are used to create HAN."

Advantages of HAN

"1. You can access this network from any corner of the house.

2. Through this, you can watch videos as well as do live streaming.

3. You can transfer data from any corner of the house.

4. All the people in the house can use this network"

Disadvantages of HAN

""1.The range of HAN network is very less, about 100 meters.

2.It takes a lot of time to transfer files and data.

3.Sometimes the server of the network also goes down.

Network Topology

"• Network topology is a structure of the network that tells how computers are connected with each other in a network.

• In other words, "Network Topology is a layout of the network, which tells how computers are connected with each other in the network and how they communicate with each other."

• Network Topology is a type of connection, which connects two or more computers together. With the help of which data and files are transferred from one computer to another.

• In network topology, internet is not required to transfer data and files. It includes network devices like switch, router and hub.

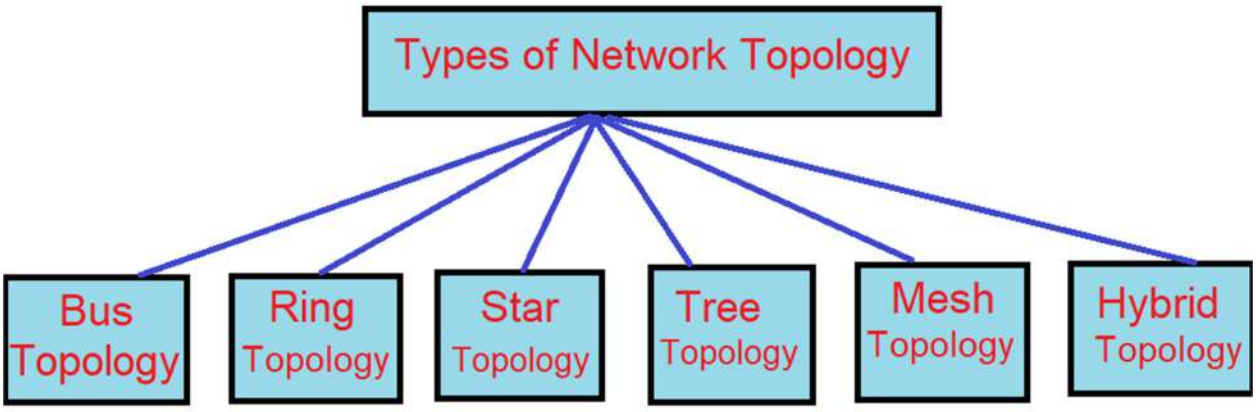
• Network topology is mostly presented in the form of a graph.

• Network topology increases the speed of data transfer.

• Network topology is defined both physically and logically."

Types of Network

"There are mainly 6 types of network topology, which are given below:-"



"• Bus topology is the simplest type of topology in which only one bus or channel is used for communication.

• In this topology, only one cable is used to connect the computers.

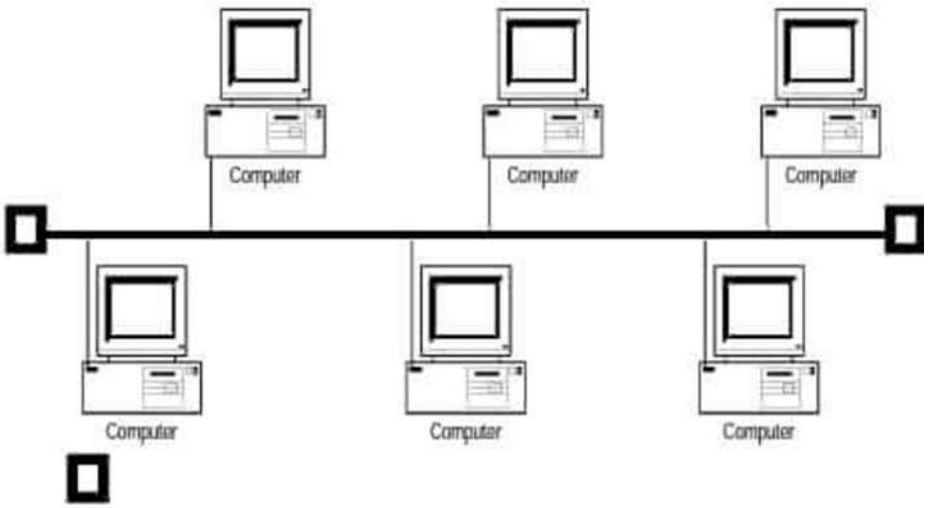
• It transfers data only in one direction, it cannot transfer data in both directions.

• Bus topology is most commonly used in 802.3 and 802.4 standard networks.

• Bus topology can be configured easily as compared to other topologies.

• Bus topology is used in small networks.

• In bus topology, the cable is also known as backbone cable. Through which data is transferred to all the computers.



Advantages of Bus Topology

"1- It costs less to set up this topology.

2- It is very easy to connect computers to the network using this topology.

3- The speed of data transfer is much better.

4- Its hardware components are easily available.

5- If there is any problem in one computer in this topology, then it does not affect the other computer.

6- In this topology, only one cable is used to connect computers to each other, which we call backbone cable.

7- It does not require hub and switch."

Disadvantages of Bus Topology

"1- In this topology, a large amount of cable is used.

2- It is not useful for large networks.

3- If there is any fault in the cable, then it affects all the computers.

4- In this topology, if two computers transfer data simultaneously, then they collide with each other, due to which the user may face problems while transferring data.

5- If new computers are connected in this topology, then its speed decreases.
6- There is lack of security in this topology."

2- Ring Topology

"• Ring topology is a type of network topology in which all computers are connected to each other in the shape of a ring. The last computer in it is connected to the first computer.

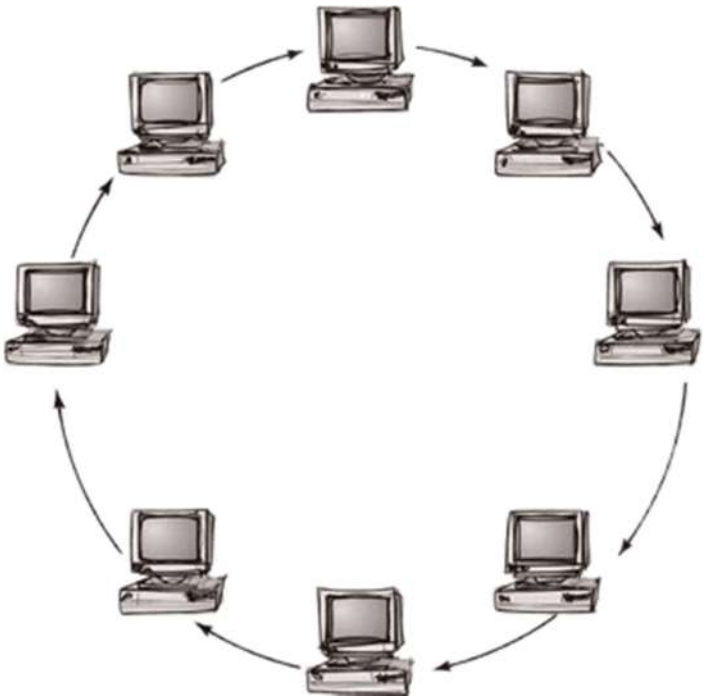
• In this topology, each computer is connected to its two nearest computers.

• This topology is like bus topology because in this data can be transferred in only one direction.

• In this topology, the data received from the previous computer is transferred forward.

• This topology has no end because in this every computer is connected to each other.

• The most common method of ring topology is token passing."



Advantages of Ring Topology

“1- Twisted pair cables are used in this topology, which are not expensive.

2- Twisted pair cables are easily available.

3- In this, data can be transferred at a very high speed.

4:- It is very easy to manage.

5:- In this, the server does not need to control the flow of data.

6- It is very easy to set up ring topology.

7- This topology can be expanded.”

Disadvantages of Ring Topology

“1- If there is a fault in one cable, the entire network fails.

2- In this topology, if one computer fails, the entire network shuts down.

3- Troubleshooting is difficult.

4:- It is difficult to add and delete new computers.

5- Adding new computers in this topology slows down the network speed.”

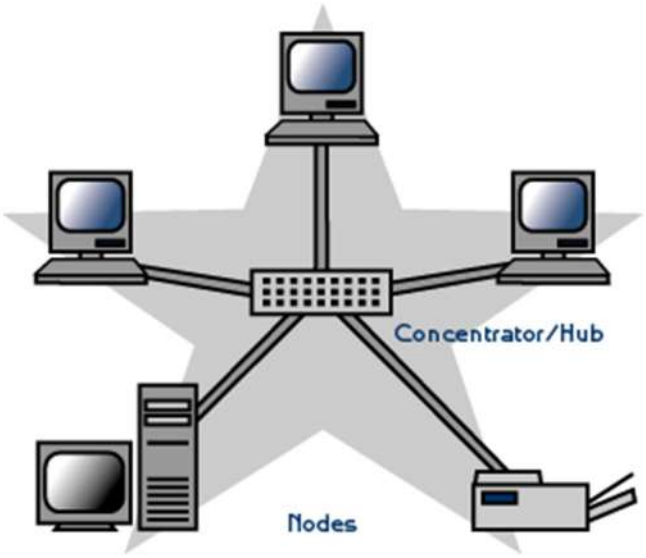
3– Star Topology

"• This is a type of network topology in which all computers are connected to a central device called a hub.

• In Star topology, the hub acts as a server and the computers act as clients.

• Star topology is also called Star Network. This is the most famous network topology.

• In Star Topology, if a computer wants to send data to another computer, it first has to send the information to the hub, then the hub transmits that data to the other computer."



Advantages of Star Topology

"1- Star topology is very easy to manage.

2- If a cable gets damaged in this topology, it does not have any negative impact on the entire network.

3- It is easy to set it up.

4- It is easy to add a new computer in it.

5- There is no collision of data in it.

6- It is easy to detect faults in it.

7- The speed of transferring data in star topology is very high.”

Disadvantages of Star Topology

“1- Star topology costs more to install.

2- It requires more cables.

3- If the hub gets damaged, all the computers stop working.

4- The hub requires more resources.”

4– Tree topology

“• Tree topology is a network topology which has a tree like structure and in it computers are connected like branches of a tree.

• Tree topology is also called hierarchical topology.

• This topology has the features of both star and bus topology.

• In tree topology the topmost computer is called root computer, and all other computers are children of the root device.

• This topology is used in Wide Area Network (WAN).

• It has parent-child hierarchy.

• In tree topology only one connection can be established between two computers.”

Advantages of Tree Topology

“1- It is very easy to manage and maintain the network.

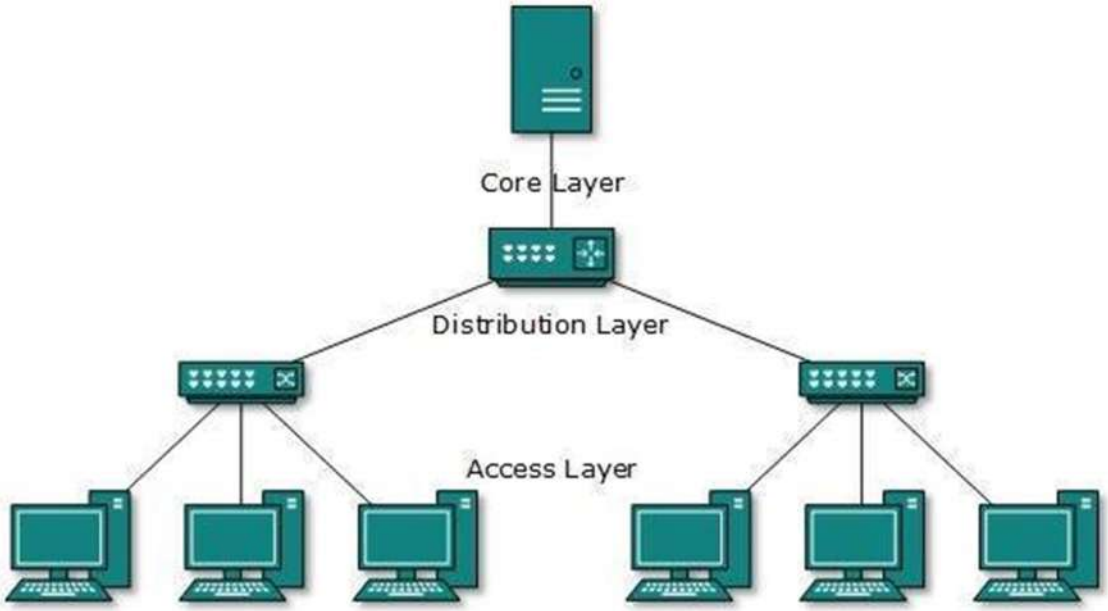
2- It is easy to detect and correct errors.

3- It is very easy to expand this topology.

4- It is very secure.

5- Tree topology is very reliable.

6- If one computer gets damaged, it does not affect the network.”



Disadvantages of Tree Topology

“1- In this topology, if any problem occurs in a computer, then it is very difficult to solve it.

2- Installing this topology costs more.

3- It is difficult to configure the network in it.

4- This topology requires a large amount of cable.”

5– Mesh topology

“• In Mesh topology, each computer is connected to another computer through a special channel.

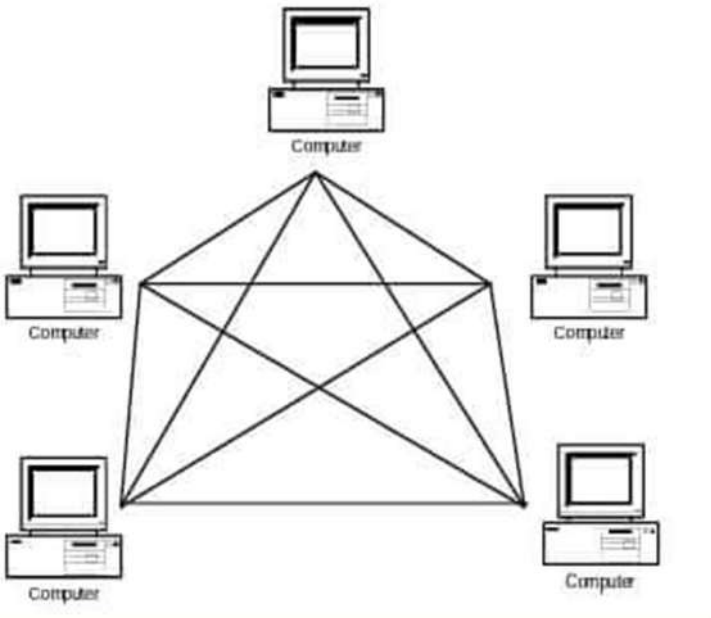
• In this, two computers can communicate directly with each other.

• This topology is mostly used in wireless networks.

• There are two types of Mesh topology:- full mesh and partial mesh.

Full mesh – In this, every computer is connected to every other computer in the network.

Partial mesh – In this, some computers are not connected to all the computers.”



Advantages of Mesh Topology

“1- This topology is very reliable.

2- It can manage high traffic as well.

3- It is very secure.

4- If any computer in it gets damaged, it does not affect the entire network.

5- It is easy to communicate in this topology.

6- The speed of transferring data in this topology is very fast."

Disadvantages of Mesh Topology

1- It is very difficult to maintain Mesh topology.

2- It is difficult to manage this topology.

3- It is very expensive.

4- Its process is very complex.

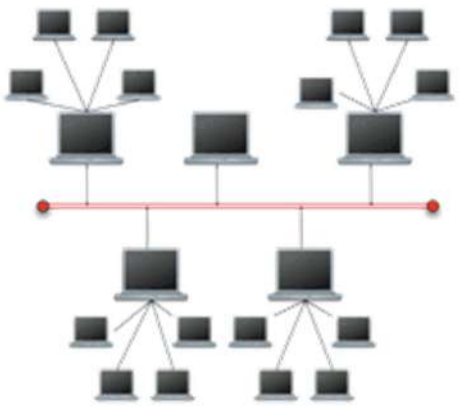
6– Hybrid Topology

"• Hybrid topology is a type of network topology that is made up of bus topology, mesh topology, ring topology, star topology and tree topology.

• It includes the features of all other topologies.

• In other words, "Hybrid topology is a network topology that is made up of two or more topologies."

• It is used in schools, businesses and other places as per the need."



Advantages of Hybrid Topology

“1- This topology is very reliable.

2- It is easy to detect errors in it.

3- It can easily handle high traffic.

4- It is used to create very large networks.

5- If computers are added in this topology, then its speed does not slow down.

6- This topology is flexible and can be designed as per the requirements.”

Disadvantages of Hybrid Topology

Its disadvantages are as follows:-

1- The structure of this topology is quite complex.

2- It is quite expensive.

3- More cables are required to install this topology, due to which it becomes even more expensive.

4- It is quite difficult to install it.

Distributed Processing

"Definition:

When a processing task is divided among several computers or processors in a computer network and they complete the task together, it is called distributed processing. In this, a large task is divided into smaller parts and processed simultaneously on different computer systems."

Key Features:

"1. Task Sharing:
The work is divided into smaller parts and run on different machines.

2. Better Resource Utilization:
The processing capacity, storage and memory of different computers in the network are used effectively.

3. Parallel Processing:
Many computers work together which makes the work complete faster.

4. Reliability:
If one computer fails, the work can continue from other computers."

Advantages

"• Faster processing speed.

• Flexibility and scalability of the system increases.

• There is not much load on a single computer.

• Shared use of resources."

Disadvantages:

"• The complexity of the network increases.

• Managing and securing data can be difficult.

• Coordination between computers is challenging."

Examples:

"• Online banking system – where different servers handle different services.

• Google search engine – millions of servers process data simultaneously.

• Cloud computing – where many servers provide services together."

"Standards and Organizations in Computer Networks

Standards and Organizations play a very important role in computer networking. Different devices, protocols and technologies of the network can work together only when they follow a common standard. The work of creating and maintaining these standards is done by different international organizations."

Standards

"Standards in networking means the rules, protocols and guidelines that determine

- how data is transferred over a network,
- how network devices communicate with each other,
- how hardware and software are interoperable."

Example

IEEE-Institute of Electrical and Electronics Engineers

- IEEE 802.3 → Standard for Ethernet
- IEEE 802.11 → Standard for Wi-Fi
- TCP/IP Protocol Suite → Standard for Internet

Definition of Ethernet:

Ethernet is a networking technology that enables data transfer by connecting computers and other devices. It is mainly used in wired networks and data is transferred in the form of packets.

Features of Ethernet:

Standard: Based on IEEE 802.3.

Topology: Earlier bus topology was used, now star topology is prevalent.

Speed: Initially 10 Mbps, now up to 100 Mbps, 1 Gbps, 10 Gbps.

Transmission Medium: Twisted pair cable, co-axial cable and optical fiber.

Types of Ethernet:

Standard Ethernet (10 Mbps)

Fast Ethernet (100 Mbps)

Gigabit Ethernet (1 Gbps)

10 Gigabit Ethernet (10 Gbps)

 In simple terms, Ethernet is a technology that exchanges data between computers and network devices by connecting them through cables."

Organizations

"The major organizations that create and enforce networking standards are as follows:

1. ISO (International Organization for Standardization)

o Defined the OSI (Open Systems Interconnection) model.

o Prepares international standards for networking.

2. IEEE (Institute of Electrical and Electronics Engineers)

o Develops networking standards such as the IEEE 802 series (Ethernet, Wi-Fi, Bluetooth, etc.).

3. IETF (Internet Engineering Task Force)

o Defines and maintains Internet protocols and standards such as TCP/IP, HTTP, SMTP.

4. ANSI (American National Standards Institute)

o Prepares standards related to networking and IT in the US.

5. ITU-T (International Telecommunication Union – Telecommunication Standardization Sector)

o Creates standards for telecommunications and network services globally.

o Standards for ISDN, DSL, and mobile networks.

6. W3C (World Wide Web Consortium)
o Develops standards for web technologies such as HTML, XML, CSS etc."

Importance of Standards & Organizations

"• Helps to connect network devices of different companies and countries.

• Ensuring interoperability.

• Ease in development and adoption of new technology.

• Maintaining uniformity in networking at global level."

Line Configuration in Computer Network

"Line configuration in a network means —

How two or more devices (such as computers, printers, servers) are connected to each other and through which route the data will flow.

This is a fundamental concept in data communication."

Types of line configurations

1. Point-to-Point Configuration

• There is a direct connection between two devices.

• Only those two devices can communicate at a time.

• Data transfer is fast and secure.

 Example:

Connection between computer and modem, telephone line.

Advantages:

• Simple and reliable.

• High speed.

Disadvantages:

• Cost increases when more devices are added.”

---

2. Multipoint Configuration / Multi-drop Line Configuration

“• In this, many devices are connected to the same communication line.

• All devices share the same channel.

• Only one device can send data at any time.”

Example:
Bus topology, shared line in LAN.

Advantages:

• Low cost.
• Easy installation.

Disadvantages:

• If one device fails, others may be affected.
• Data transfer speed may decrease.
• Problem of data collision.”

“Conclusion:

• Point-to-Point → Suitable for fewer devices and faster communication.

• Multipoint → Economical for connecting multiple devices together.”

OSI Model

"• The full name of OSI model is Open System Interconnection. It was developed by ISO (International Organization for Standardization) in 1984 and this model has 7 layers.

• OSI model is a reference model for communication between two users in a network. Each layer of this model does not depend on the other layer but data is transmitted from one layer to another.

• OSI model is a reference model, that is, it is not used in real life but it is used only as a reference.

• OSI model describes how data or information is sent and received in a network. All the layers of this model have their own different work so that data can reach from one system to another easily.

• OSI model also describes how network hardware and software work with each other as layers. • This model has 7 layers and each layer has its own specific function."

7 Layers of OSI MODEL

The OSI model has the following 7 layers

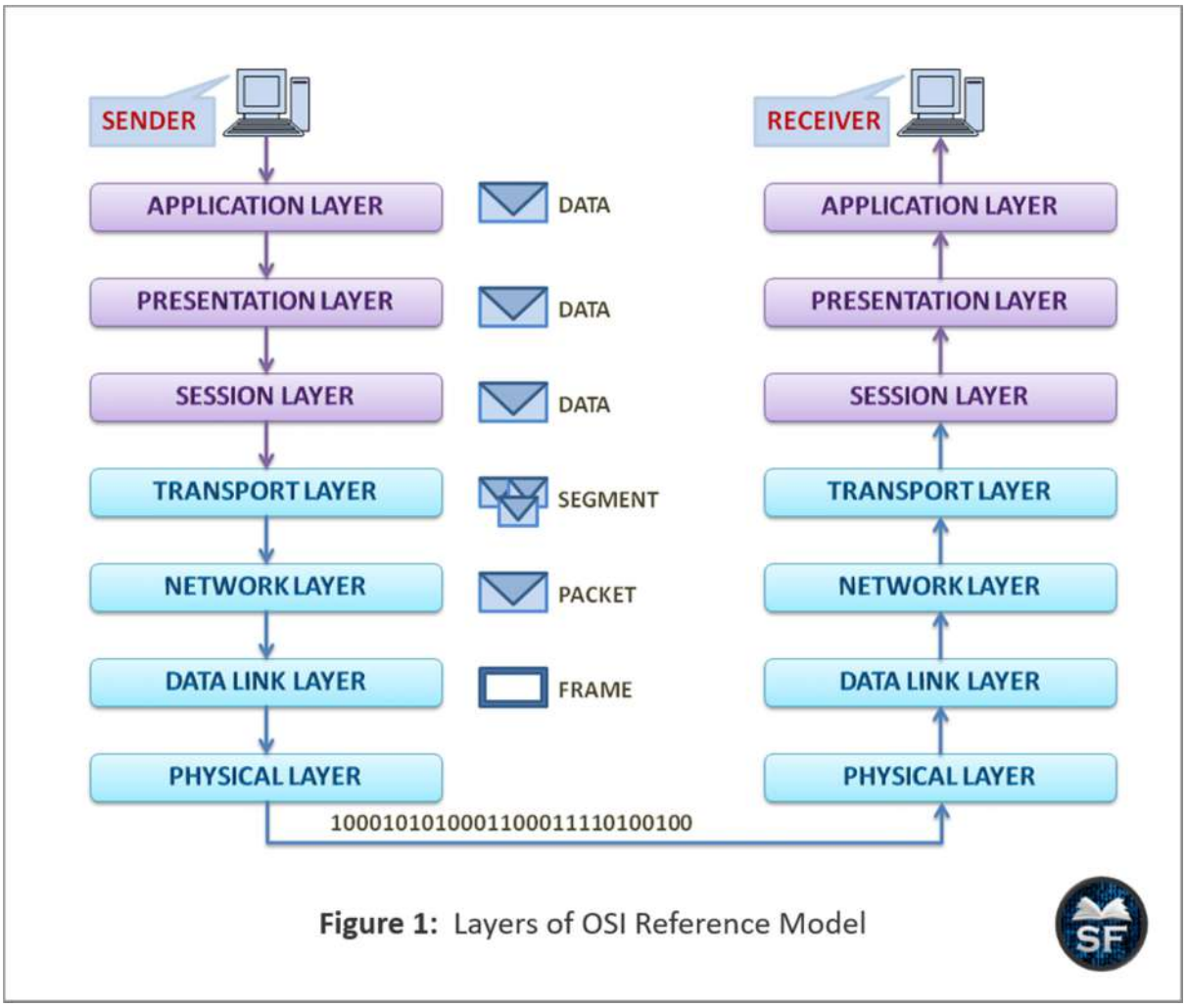
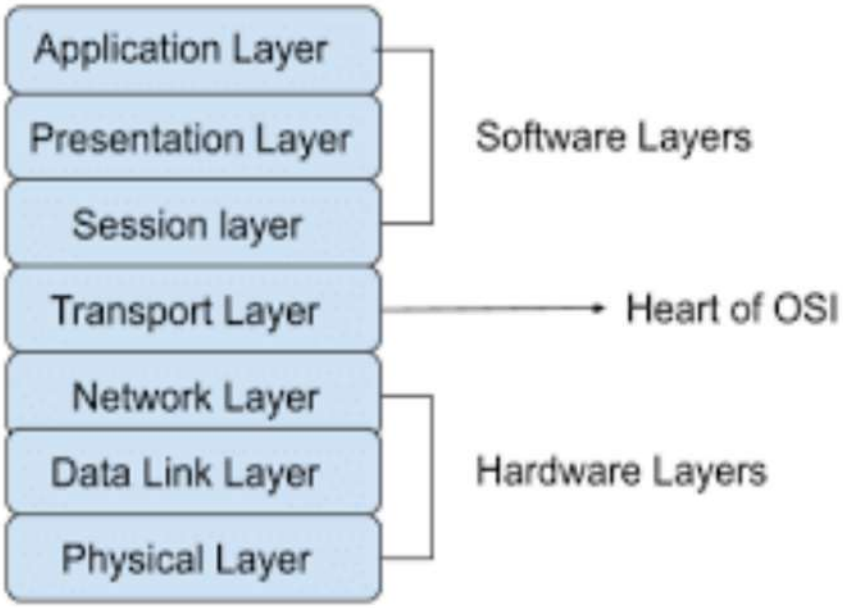
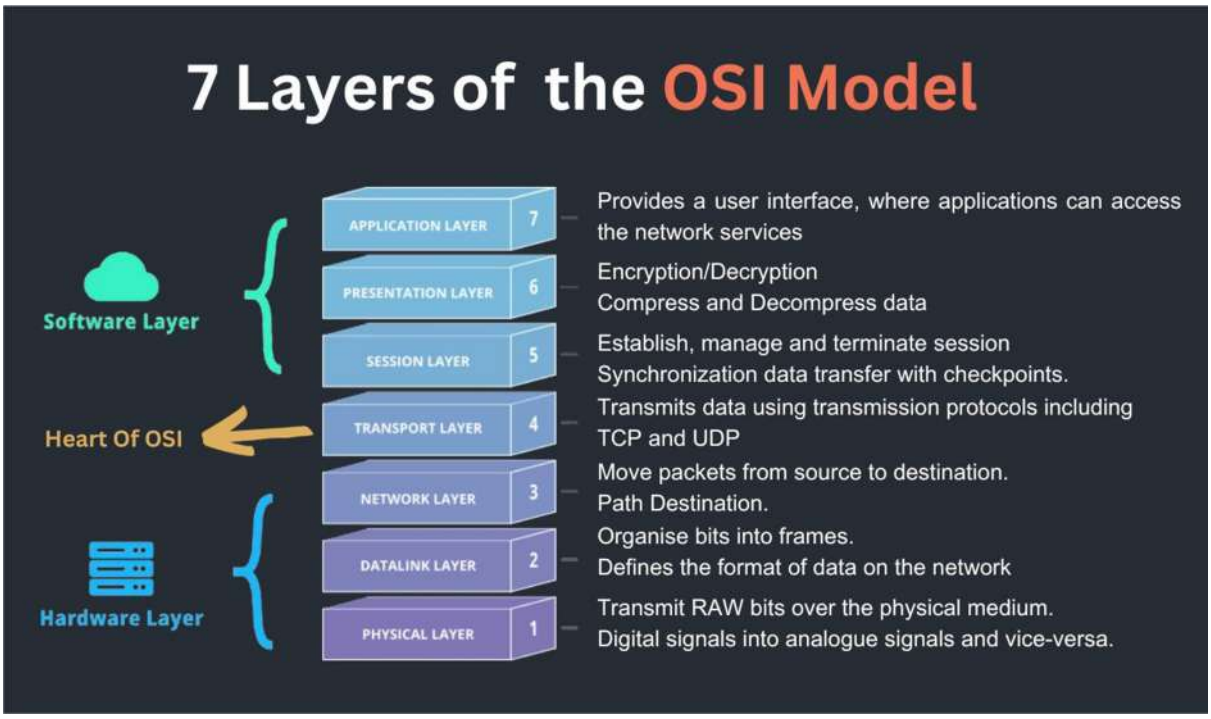


Figure 1: Layers of OSI Reference Model

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OSI Model in Computer Network





OSI Model all Layer

1. Physical Layer Data Transmission Media (Wired/Wireless)

- Signal Encoding
- Transmission Rate

"• This is the lowest layer of the OSI model.

• Transmits data in the form of bits (0s and 1s).

• This layer controls the network hardware and media (cables, switches, hubs, NIC).

• Functions:
o Data Transmission Media (Wired/Wireless)
o Signal Encoding
o Determine Transmission Rate"

2. Data Link Layer

“• It manages data in the form of frames.

• This layer creates a reliable link between two devices.

• Functions:

o Error Detection and Error Correction

o MAC Addressing (Physical Address)

o Flow Control

• Devices: Switch, Bridge”

---

3. Network Layer ()

“• Breaks data into packets and sends it from source to destination.

• Function:

o Logical Addressing (IP Address)

o Routing (choosing the right path)

o Congestion Control

• Device: Router

• Protocol: IP, ICMP”

---

4. Transport Layer

“• It divides data into segments.

• It ensures End-to-End Communication.

• Functions:

o Reliable Delivery (TCP) or Fast Delivery (UDP)

o Error Recovery

o Flow Control

• Protocols: TCP, UDP”

---

5. Session Layer

“• This layer creates, manages and terminates the communication session between the sender and receiver.

• Functions:

o Session Establishment, Maintenance, Termination
o Synchronization (Creating Checkpoints)

• Example: Remote Procedure Call (RPC), NetBIOS”

---

6. Presentation Layer

“• It presents data in a format understandable to the user.

• Functions:
o Data Translation (ASCII ↔ EBCDIC, JPEG, MP4 etc.)
o Data Compression
o Encryption & Decryption (Security)”

---

7. Application Layer

“• It is the topmost layer of the OSI model and is directly closest to the user.

• Functions:
o Provide network services to the user
o Email, File Transfer, Web Browsing

• Protocols: HTTP, FTP, SMTP, DNS, Telnet”

“Telnet (Telecommunication Network) is a network protocol used to connect to and control a remote computer or network device.

This protocol was developed in 1973 and operates at the Application Layer of the OSI model.”

“Features of Telnet

1. Provides Remote Login Service – meaning a user can log in to another computer/server from their computer.

2. It provides a Command Line Interface (CLI).

3. Telnet's default port number is 23.

4. Works on a client-server model (Telnet Client ↔ Telnet Server).

5. It sends data in plain text (no encryption, so it is insecure).”

“Uses of Telnet

• To remotely configure a router, switch, or server.

• To log in to a remote server.

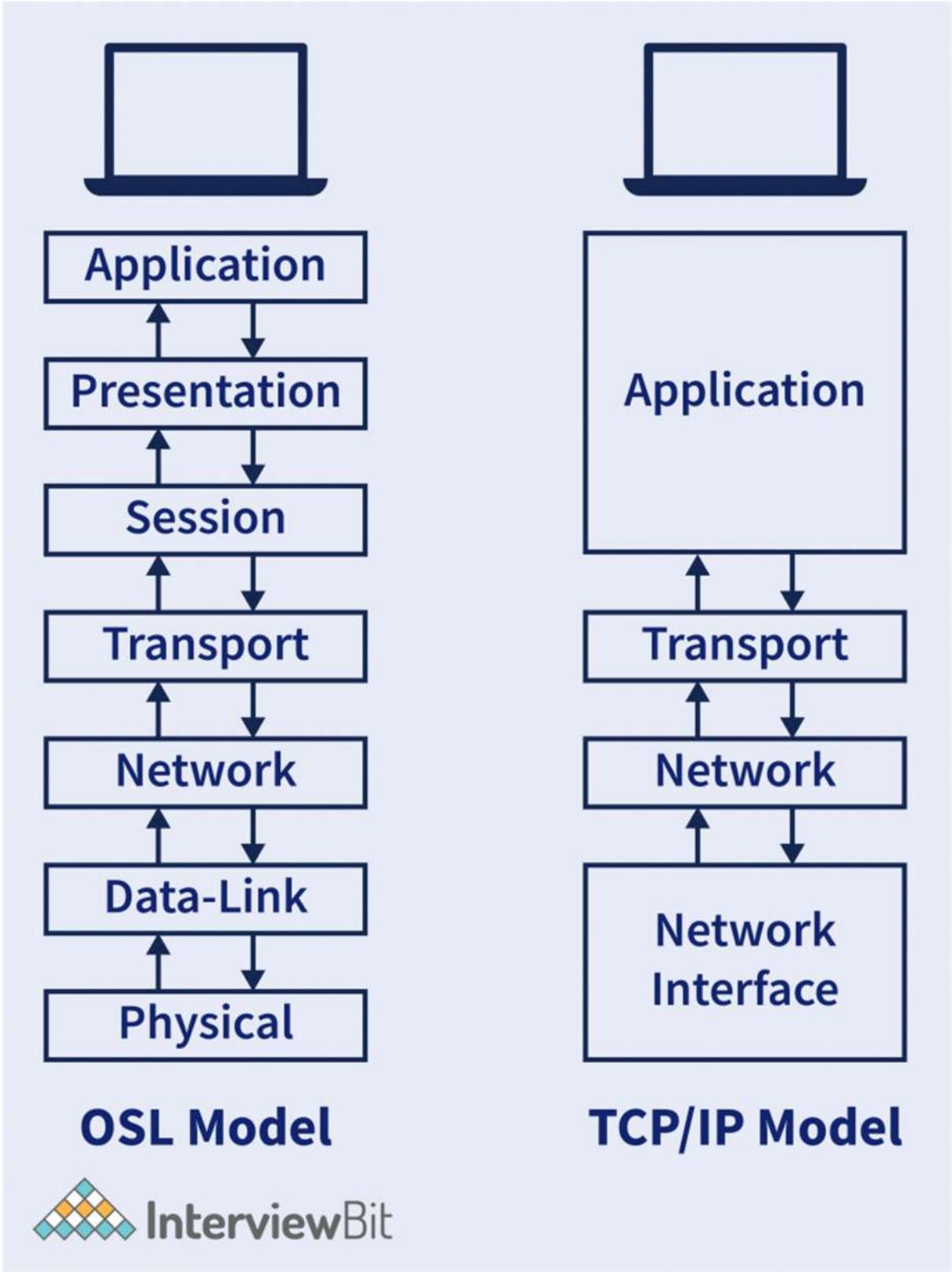
• To access Internet services (such as library catalogs, BBSes) in the past.”

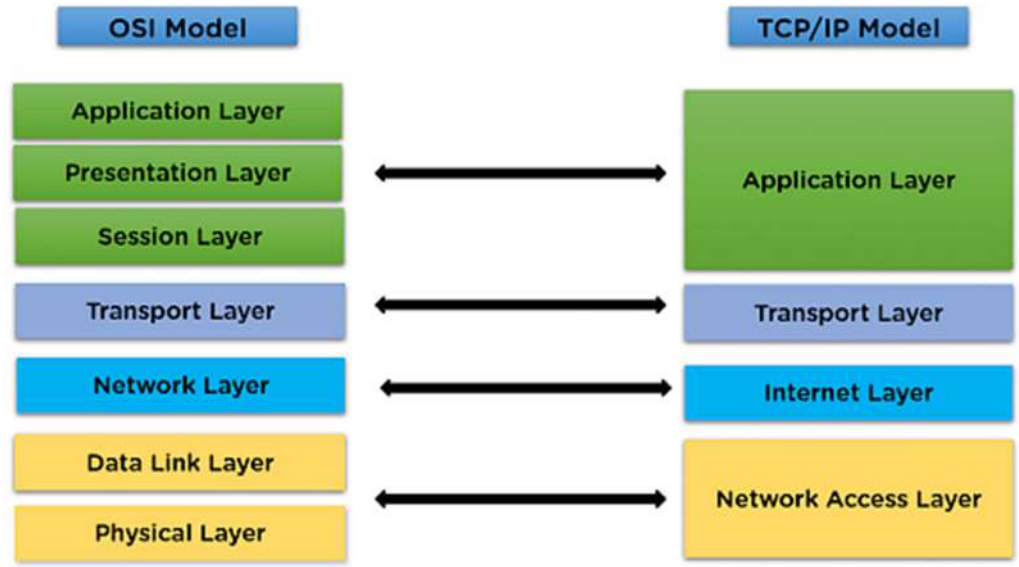
“**Limitations of Telnet**

• It lacks security (data and passwords are sent without encryption).

• Nowadays, SSH (Secure Shell) is more commonly used in its place because SSH is encrypted.”

”


Application
Presentation
Session
Transport
Network
Data-Link
Physical
OSL Model
Application
Transport
Network
Network Interface
TCP/IP Model
InterviewBit



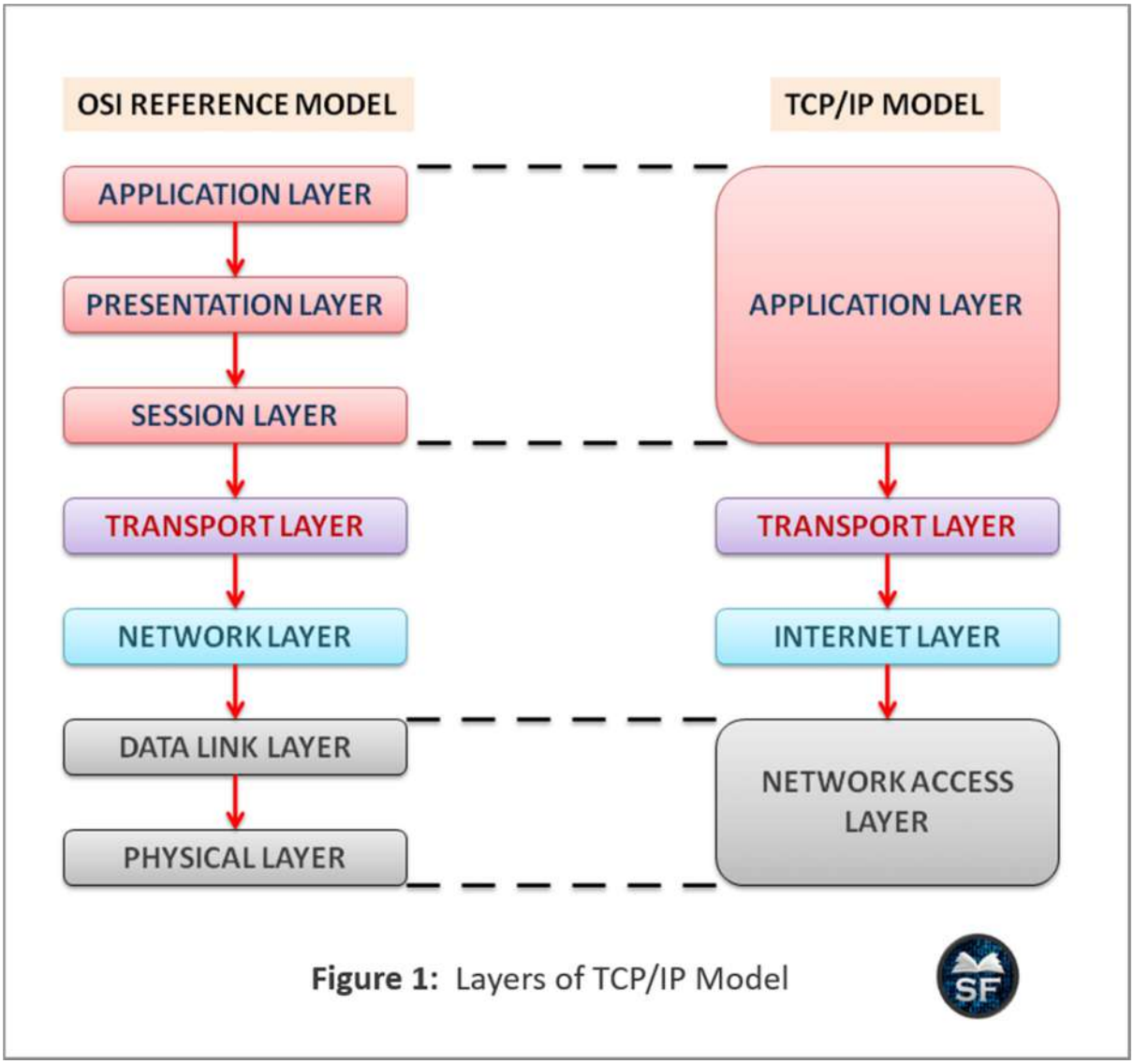


Figure 1: Layers of TCP/IP Model

TCP/IP Protocol in Computer Network

"TCP/IP (Transmission Control Protocol / Internet Protocol) is the most widely used communication protocol suite for networking. It is called the backbone of the Internet.

 It was developed by DARPA (Defense Advanced Research Projects Agency, USA) in the 1970s.

 TCP/IP model is also called DoD (Department of Defense Model)."

TCP/IP Model

"The TCP/IP model has 4 layers (not 7 like OSI)."

1. Application Layer

"• This layer provides network services to the user.

• Various protocols work here which are used according to the application.

• Example:

o HTTP, HTTPS → Web Browsing
o SMTP, POP3, IMAP → Email
o FTP, SFTP → File Transfer
o DNS → Converting name to IP Address"

The Full Form of HTTPS is HyperText Transfer Protocol Secure.
SMTP stands for Simple Mail Transfer Protocol
IMAP stands for Internet Message Access Protocol
The full form of POP3 is Post Office Protocol version 3.
SFTP stands for either Secure File Transfer Protocol
The Domain Name System (DNS)

---

2. Transport Layer

"• It handles End-to-End Communication and Reliability.

• It sends data by breaking it into small segments.

• Protocol:

o TCP (Transmission Control Protocol) → Reliable, Connection-Oriented
o UDP (User Datagram Protocol) → Fast, Connectionless”

---

3. Internet Layer (

“• It manages data in packets.

• Routing is done to send data from source to destination.

• Protocol:

o IP (Internet Protocol) → Logical Addressing (IPv4/IPv6)

o ICMP (Internet Control Message Protocol) → Error Reporting (eg Ping)

o ARP (Address Resolution Protocol) → Convert IP to MAC Address

o RARP (Reverse ARP) → Convert MAC to IP”

---

4. Network Access Layer (or Link Layer)

“• It works as Data Link + Physical Layer of OSI.

• It sends data in the form of frames and bits at hardware level.

• Example: Ethernet, Wi-Fi, Token Ring”

Difference between OSI & TCP/IP

"• OSI has 7 layers while TCP/IP has 4.

• TCP/IP is a more practical and implemented model.

• All communication on the Internet takes place via TCP/IP."

"

Differences between OSI Model and TCP/IP Model

Parameters	OSI Model	TCP/IP Model
Full Form	OSI stands for Open Systems Interconnection.	TCP/IP stands for Transmission Control Protocol/Internet Protocol.
Layers	It has 7 layers.	It has 4 layers.
Usage	It is low in usage.	It is mostly used.
Approach	It is vertically approached.	It is horizontally approached.
Delivery	Delivery of the package is guaranteed in OSI Model.	Delivery of the package is not guaranteed in TCP/IP Model.
Replacement	Replacement of tools and changes can easily be done in this model.	Replacing the tools is not easy as it is in OSI Model.
Reliability	It is less reliable than TCP/IP Model.	It is more reliable than OSI Model.

In short:”

“• OSI Model → It is a theoretical model, which helps in understanding networking and creating standards.”

“• TCP/IP Model → It is a practical model, which is used in real internet and networking.”

Repeater

Repeaters boost the strength of a signal as it travels through a communication channel, like a telephone line or a radio frequency, to help it cover longer distances.

In situations where signals lose power due to resistance or distance, a repeater amplifies the signal before sending it further.

In computer networking, repeaters operate on the physical layer of the OSI model, as they only work with the actual signal and don't interpret data.

•

Repeaters are used to extend transmissions so that the signal can cover longer distances or be received on the other side of an obstruction.

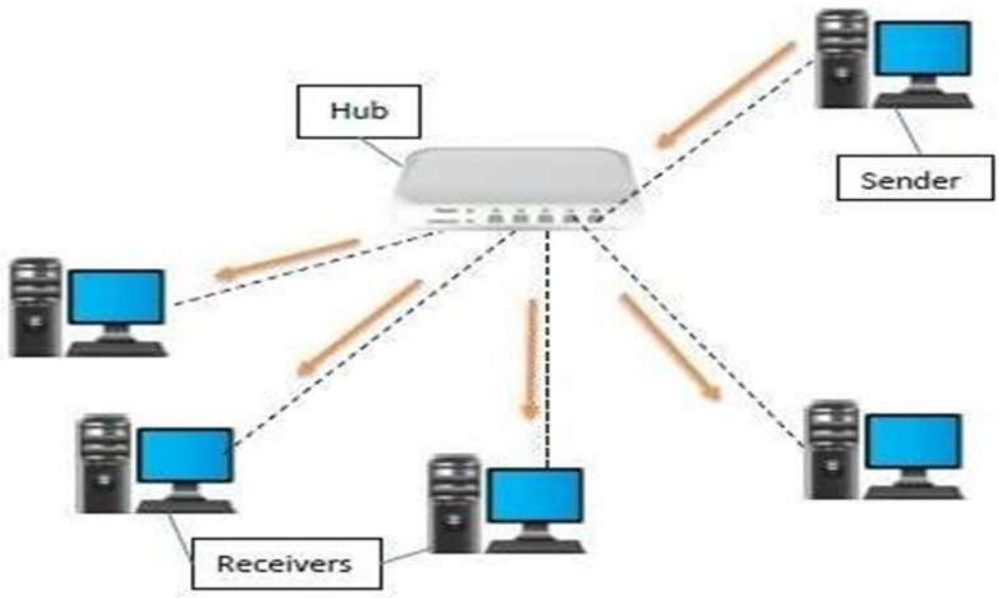
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In computer networking, because repeaters work with the actual physical signal, and do not attempt to interpret the data being transmitted, they operate on the physical layer, the first layer of theOSI model.

Hubs

A hub is a **physical layer** networking device which is used to connect multiple devices in a network. They are generally used to connect computers in a **LAN**.

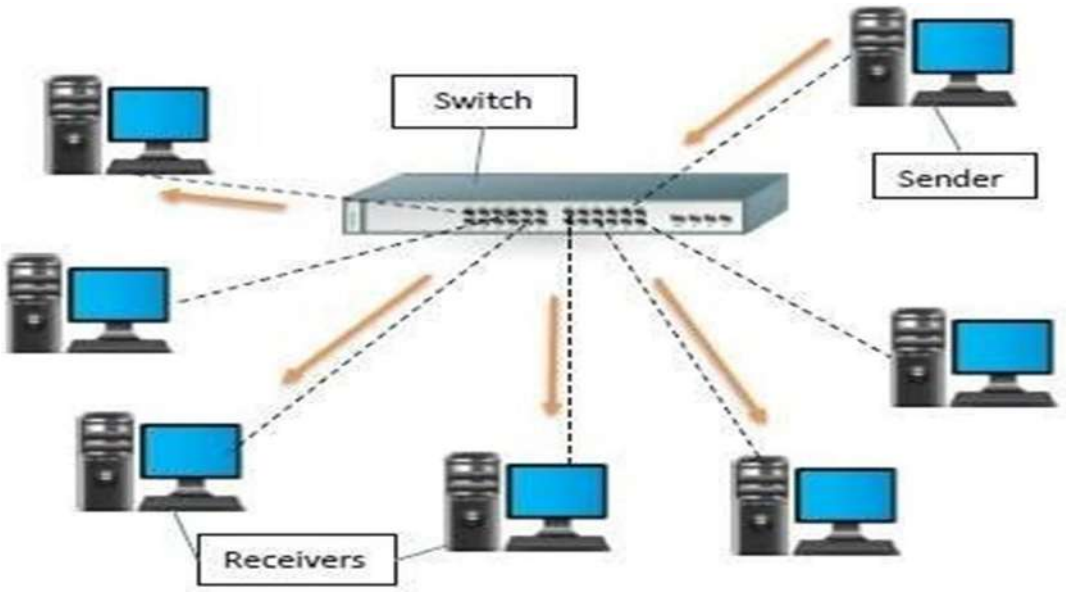
A hub has many ports in it. A computer which intends to be connected to the network is plugged in to one of these ports. When a data frame arrives at a port, it is broadcast to every other port, without considering whether it is destined for a particular destination or not.



Switches

A switch is a **data link layer** networking device which connects devices in a network and uses **packet switching** to send and receive data over the network.

Like a hub, a switch also has many ports, to which computers are plugged in. However, when a data frame arrives at any port of a **network switch**, it examines the destination address and sends the frame to the corresponding device(s). Thus, it supports both unicast and **multicast communications**.



Differences between Hub and Switch

Hub	*Switch*
They operate in the physical layer of the OSI model.	They operate in the data link layer of the OSI model.
It is a non-intelligent network device that sends message to all ports.	It is an intelligent network device that sends message to selected destination ports.
It primarily broadcasts messages.	It is supports unicast, multicast and broadcast.
Transmission mode is half duplex.	Transmission mode is full duplex.
Collisions may occurs during setup of transmission when more than one computers place data simultaneously in the corresponding ports.	Collisions do not occur since the communication is full duplex.

Hub	*Switch*
They are passive devices, they don't have any software associated with it.	They are active devices, equipped with network software.
They generally have fewer ports of 4/12.	The number of ports is higher – 24/48.

A **bridge** is a hardware or software device used in computer networking that joins two or more network segments and routes data between them according to MAC addresses. In short, a bridge improves network performance and connection by functioning at the data link layer (Layer 2) of the OSI model.

